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Reload shaft assembly for surgical stapler

Abstract

A surgical stapling system can include a reload shaft. The shaft can include an elongate tubular member with have a jaw assembly at the distal end thereof and a coupling collar at the proximal end thereof. The shaft assembly also includes an articulation joint coupling the jaw assembly to the distal end. A drive member and an articulation member extend within the tubular body of the shaft from the proximal end to the distal end. A firing member is connected to the distal end of the drive member such that advancement of the drive beam advances the firing member to close the jaw assemblies and fire staples from a reload positioned in the jaw assembly. The shaft assembly can also include a lockout mechanism to prevent a firing operation on a previously-fired reload or no reload.

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7784663	12/2009	Shelton, IV	N/A	N/A
7793812	12/2009	Moore et al.	N/A	N/A
7798386	12/2009	Schall et al.	N/A	N/A
7810693	12/2009	Broehl et al.	N/A	N/A
7815090	12/2009	Marczyk	N/A	N/A
7815091	12/2009	Marczyk	N/A	N/A
7819298	12/2009	Hall et al.	N/A	N/A
7819896	12/2009	Racenet	N/A	N/A
7823760	12/2009	Zemlok et al.	N/A	N/A
7828188	12/2009	Jankowski	N/A	N/A
7828189	12/2009	Holsten et al.	N/A	N/A
7837079	12/2009	Holsten et al.	N/A	N/A
7837081	12/2009	Holsten et al.	N/A	N/A
7845534	12/2009	Viola et al.	N/A	N/A
7845535	12/2009	Scircia	N/A	N/A
7845537	12/2009	Shelton, IV et al.	N/A	N/A
7857184	12/2009	Viola	N/A	N/A
7857185	12/2009	Swayze et al.	N/A	N/A
7857187	12/2009	Milliman	N/A	N/A
7861906	12/2010	Doll et al.	N/A	N/A
7866525	12/2010	Scirica	N/A	N/A
7866527	12/2010	Hall et al.	N/A	N/A
7891534	12/2010	Wenchell et al.	N/A	N/A
7905381	12/2010	Baxter, III et al.	N/A	N/A
7909220	12/2010	Viola	N/A	N/A
7909221	12/2010	Viola et al.	N/A	N/A
7913891	12/2010	Doll et al.	N/A	N/A
7914543	12/2010	Roth et al.	N/A	N/A
7918230	12/2010	Whitman et al.	N/A	N/A
7918376	12/2010	Knodel et al.	N/A	N/A
7918377	12/2010	Measamer et al.	N/A	N/A
7922063	12/2010	Zemlok et al.	N/A	N/A
7934628	12/2010	Wenchell et al.	N/A	N/A
7934629	12/2010	Wixey et al.	N/A	N/A
7934630	12/2010	Shelton, IV et al.	N/A	N/A
7942300	12/2010	Rethy et al.	N/A	N/A
7954685	12/2010	Viola	N/A	N/A
7954686	12/2010	Baxter, III et al.	N/A	N/A
7959050	12/2010	Smith et al.	N/A	N/A
7963433	12/2010	Whitman et al.	N/A	N/A
7992758	12/2010	Whitman et al.	N/A	N/A
8002795	12/2010	Beetel	N/A	N/A
8006887	12/2010	Marczyk	N/A	N/A
8007513	12/2010	Nalagatla et al.	N/A	N/A
8008598	12/2010	Whitman et al.	N/A	N/A
8011550	12/2010	Aranyi et al.	N/A	N/A
8011553	12/2010	Mastri et al.	N/A	N/A
8012170	12/2010	Whitman et al.	N/A	N/A
8016178	12/2010	Olson et al.	N/A	N/A
8020742	12/2010	Marczyk	N/A	N/A

8020743	12/2010	Shelton, IV	N/A	N/A
8028885	12/2010	Smith et al.	N/A	N/A
8033438	12/2010	Scirica	N/A	N/A
8033440	12/2010	Wenchell et al.	N/A	N/A
8033441	12/2010	Marczyk	N/A	N/A
8033442	12/2010	Racenet et al.	N/A	N/A
8034077	12/2010	Smith et al.	N/A	N/A
8038046	12/2010	Smith et al.	N/A	N/A
8052024	12/2010	Viola et al.	N/A	N/A
8056788	12/2010	Mastri et al.	N/A	N/A
8056789	12/2010	White et al.	N/A	N/A
8061576	12/2010	Cappola	N/A	N/A
8061577	12/2010	Racenet et al.	N/A	N/A
8070033	12/2010	Milliman et al.	N/A	N/A
8070034	12/2010	Knodel	N/A	N/A
8070035	12/2010	Holsten et al.	N/A	N/A
8070036	12/2010	Knodel	N/A	N/A
8074861	12/2010	Ehrenfels et al.	N/A	N/A
8083118	12/2010	Milliman et al.	N/A	N/A
8087563	12/2011	Milliman et al.	N/A	N/A
8091753	12/2011	Viola	N/A	N/A
8091754	12/2011	Ehrenfels et al.	N/A	N/A
8092493	12/2011	Marczyk	N/A	N/A
8100309	12/2011	Marczyk	N/A	N/A
8113406	12/2011	Holsten et al.	N/A	N/A
8113407	12/2011	Holsten et al.	N/A	N/A
8113408	12/2011	Wenchell et al.	N/A	N/A
8113410	12/2011	Hall et al.	N/A	N/A
8118207	12/2011	Racenet et al.	N/A	N/A
8123100	12/2011	Holsten et al.	N/A	N/A
8127976	12/2011	Scirica et al.	N/A	N/A
8136712	12/2011	Zingman	N/A	N/A
8152041	12/2011	Kostrzewski	N/A	N/A
8157145	12/2011	Shelton, IV et al.	N/A	N/A
8157150	12/2011	Viola et al.	N/A	N/A
8157152	12/2011	Holsten et al.	N/A	N/A
8181839	12/2011	Beetel	N/A	N/A
8186555	12/2011	Shelton, IV et al.	N/A	N/A
8186556	12/2011	Viola	N/A	N/A
8186560	12/2011	Hess et al.	N/A	N/A
8191752	12/2011	Scirica	N/A	N/A
8196795	12/2011	Moore et al.	N/A	N/A
8201721	12/2011	Zemlok et al.	N/A	N/A
8205619	12/2011	Shah et al.	N/A	N/A
8205780	12/2011	Sorrentino et al.	N/A	N/A
8205781	12/2011	Baxter, III et al.	N/A	N/A
8210411	12/2011	Yates et al.	N/A	N/A
8210416	12/2011	Milliman et al.	N/A	N/A
8220688	12/2011	Laurent et al.	N/A	N/A
8225979	12/2011	Farascioni et al.	N/A	N/A

8231040	12/2011	Zemlok et al.	N/A	N/A
8231041	12/2011	Marczyk et al.	N/A	N/A
8235274	12/2011	Cappola	N/A	N/A
8236010	12/2011	Ortiz et al.	N/A	N/A
8240536	12/2011	Marczyk	N/A	N/A
8240537	12/2011	Marczyk	N/A	N/A
8241322	12/2011	Whitman et al.	N/A	N/A
8245898	12/2011	Smith et al.	N/A	N/A
8245899	12/2011	Swensgard et al.	N/A	N/A
8245900	12/2011	Scirica	N/A	N/A
8256656	12/2011	Milliman et al.	N/A	N/A
8272552	12/2011	Holsten et al.	N/A	N/A
8272554	12/2011	Whitman et al.	N/A	N/A
8281972	12/2011	Wixey et al.	N/A	N/A
8281973	12/2011	Wenchell et al.	N/A	N/A
8286846	12/2011	Smith et al.	N/A	N/A
8292146	12/2011	Holsten et al.	N/A	N/A
8292148	12/2011	Viola	N/A	N/A
8292151	12/2011	Viola	N/A	N/A
8292152	12/2011	Milliman et al.	N/A	N/A
8292153	12/2011	Jankowski	N/A	N/A
8292157	12/2011	Smith et al.	N/A	N/A
8308041	12/2011	Kostrzewski	N/A	N/A
8308043	12/2011	Bindra et al.	N/A	N/A
8317070	12/2011	Hueil et al.	N/A	N/A
8322455	12/2011	Shelton, IV et al.	N/A	N/A
8336754	12/2011	Cappola et al.	N/A	N/A
8342377	12/2012	Milliman et al.	N/A	N/A
8342378	12/2012	Marczyk et al.	N/A	N/A
8342379	12/2012	Whitman et al.	N/A	N/A
8342380	12/2012	Viola	N/A	N/A
8348125	12/2012	Viola et al.	N/A	N/A
8348129	12/2012	Bedi et al.	N/A	N/A
8348131	12/2012	Omaits et al.	N/A	N/A
8353440	12/2012	Whitman et al.	N/A	N/A
8360297	12/2012	Shelton, IV et al.	N/A	N/A
8360299	12/2012	Zemlok et al.	N/A	N/A
8393513	12/2012	Jankowski	N/A	N/A
8397972	12/2012	Kostrzewski	N/A	N/A
8397973	12/2012	Hausen	N/A	N/A
8403198	12/2012	Sorrentino et al.	N/A	N/A
8413868	12/2012	Cappola	N/A	N/A
8414577	12/2012	Boudreaux et al.	N/A	N/A
8418906	12/2012	Farascioni et al.	N/A	N/A
8418907	12/2012	Johnson et al.	N/A	N/A
8418908	12/2012	Beardsley	N/A	N/A
8419768	12/2012	Marczyk	N/A	N/A
8439246	12/2012	Knodel	N/A	N/A
8444036	12/2012	Shelton, IV	N/A	N/A
8453907	12/2012	Laurent et al.	N/A	N/A

8453912	12/2012	Mastri et al.	N/A	N/A
8453913	12/2012	Milliman	N/A	N/A
8459520	12/2012	Giordano et al.	N/A	N/A
8459522	12/2012	Marczyk	N/A	N/A
8464922	12/2012	Marczyk	N/A	N/A
8469252	12/2012	Holcomb et al.	N/A	N/A
8479967	12/2012	Marczyk	N/A	N/A
8496152	12/2012	Viola	N/A	N/A
8496155	12/2012	Knodel	N/A	N/A
8496156	12/2012	Sniffin et al.	N/A	N/A
8496683	12/2012	Prommersberger et al.	N/A	N/A
8505799	12/2012	Viola et al.	N/A	N/A
8505801	12/2012	Ehrenfels et al.	N/A	N/A
8517239	12/2012	Scheib et al.	N/A	N/A
8517240	12/2012	Mata et al.	N/A	N/A
8523043	12/2012	Ullrich et al.	N/A	N/A
8540130	12/2012	Moore et al.	N/A	N/A
8540133	12/2012	Bedi et al.	N/A	N/A
8540625	12/2012	Miyoshi	N/A	N/A
8544712	12/2012	Jankowski	N/A	N/A
8556151	12/2012	Viola	N/A	N/A
8556152	12/2012	Marczyk et al.	N/A	N/A
8556153	12/2012	Knodel	N/A	N/A
8561871	12/2012	Rajappa et al.	N/A	N/A
8561874	12/2012	Scirica	N/A	N/A
8573459	12/2012	Smith et al.	N/A	N/A
8573460	12/2012	Cappola	N/A	N/A
8573462	12/2012	Smith et al.	N/A	N/A
8573463	12/2012	Scirica et al.	N/A	N/A
8573464	12/2012	Nalagatla et al.	N/A	N/A
8579176	12/2012	Smith et al.	N/A	N/A
8579177	12/2012	Beetel	N/A	N/A
8584919	12/2012	Hueil et al.	N/A	N/A
8584921	12/2012	Scirica	N/A	N/A
8596513	12/2012	Olson	N/A	N/A
8608043	12/2012	Scirica	N/A	N/A
8608045	12/2012	Smith et al.	N/A	N/A
8616427	12/2012	Viola	N/A	N/A
8622274	12/2013	Yates et al.	N/A	N/A
8627992	12/2013	Edoga et al.	N/A	N/A
8627993	12/2013	Smith et al.	N/A	N/A
8627995	12/2013	Smith et al.	N/A	N/A
8631990	12/2013	Park et al.	N/A	N/A
8632525	12/2013	Kerr et al.	N/A	N/A
8632535	12/2013	Shelton, IV et al.	N/A	N/A
8636189	12/2013	Knodel et al.	N/A	N/A
8636190	12/2013	Zemlok et al.	N/A	N/A
8636192	12/2013	Farascioni et al.	N/A	N/A
8636193	12/2013	Whitman et al.	N/A	N/A

8636762	12/2013	Whitman et al.	N/A	N/A
8636766	12/2013	Milliman et al.	N/A	N/A
8657174	12/2013	Yates et al.	N/A	N/A
8657176	12/2013	Shelton, IV et al.	N/A	N/A
8657178	12/2013	Hueil et al.	N/A	N/A
8672209	12/2013	Crainich	N/A	N/A
8672951	12/2013	Smith et al.	N/A	N/A
8685004	12/2013	Zemlock et al.	N/A	N/A
8695865	12/2013	Smith et al.	N/A	N/A
8696665	12/2013	Hunt et al.	N/A	N/A
8708211	12/2013	Zemlok et al.	N/A	N/A
8708213	12/2013	Shelton, IV et al.	N/A	N/A
8740034	12/2013	Morgan et al.	N/A	N/A
8740035	12/2013	Mastri et al.	N/A	N/A
8740036	12/2013	Williams	N/A	N/A
8752748	12/2013	Whitman et al.	N/A	N/A
8763876	12/2013	Kostrzewski	N/A	N/A
8770458	12/2013	Scirica	N/A	N/A
8770459	12/2013	Racenet et al.	N/A	N/A
8789741	12/2013	Baxter, III et al.	N/A	N/A
8800839	12/2013	Beetel	N/A	N/A
8800840	12/2013	Jankowski	N/A	N/A
8800841	12/2013	Ellerhorst et al.	N/A	N/A
8806973	12/2013	Ross et al.	N/A	N/A
8807414	12/2013	Ross et al.	N/A	N/A
8820603	12/2013	Shelton, IV et al.	N/A	N/A
8820608	12/2013	Miyamoto	N/A	N/A
8833631	12/2013	Munro, III et al.	N/A	N/A
8840003	12/2013	Morgan et al.	N/A	N/A
8858571	12/2013	Shelton, IV et al.	N/A	N/A
8875971	12/2013	Hall et al.	N/A	N/A
8875972	12/2013	Weisenburgh, II et al.	N/A	N/A
8887979	12/2013	Mastri et al.	N/A	N/A
8899462	12/2013	Kostrzewski et al.	N/A	N/A
8899463	12/2013	Schall et al.	N/A	N/A
8905288	12/2013	Wenchell	N/A	N/A
8920435	12/2013	Smith et al.	N/A	N/A
8925783	12/2014	Zemlok et al.	N/A	N/A
8931679	12/2014	Kostrzewski	N/A	N/A
8931683	12/2014	Racenet et al.	N/A	N/A
8939343	12/2014	Milliman et al.	N/A	N/A
8967444	12/2014	Beetel	N/A	N/A
8967446	12/2014	Beardsley et al.	N/A	N/A
8967447	12/2014	Hartoumbekis	N/A	N/A
8968276	12/2014	Zemlok et al.	N/A	N/A
8973803	12/2014	Hall et al.	N/A	N/A
8979827	12/2014	Cappola	N/A	N/A
9004340	12/2014	Scirica	N/A	N/A
9010611	12/2014	Ross et al.	N/A	N/A

9016541	12/2014	Viola et al.	N/A	N/A
9016545	12/2014	Aranyi et al.	N/A	N/A
9022271	12/2014	Scirica	N/A	N/A
9023014	12/2014	Chowaniec et al.	N/A	N/A
9027817	12/2014	Milliman et al.	N/A	N/A
9027818	12/2014	Scirica et al.	N/A	N/A
9033202	12/2014	Scirica	N/A	N/A
9038880	12/2014	Donohoe	N/A	N/A
9055943	12/2014	Zemlok et al.	N/A	N/A
9072515	12/2014	Hall et al.	N/A	N/A
9084601	12/2014	Moore et al.	N/A	N/A
9101358	12/2014	Kerr et al.	N/A	N/A
9161813	12/2014	Benamou	N/A	N/A
9204876	12/2014	Cappola et al.	N/A	N/A
9237890	12/2015	Kostrzewski	N/A	N/A
9265585	12/2015	Wingardner et al.	N/A	N/A
9282966	12/2015	Shelton, IV et al.	N/A	N/A
9386984	12/2015	Aronhalt et al.	N/A	N/A
9402629	12/2015	Ehrenfels et al.	N/A	N/A
9510830	12/2015	Shelton, IV et al.	N/A	N/A
9532782	12/2016	Kostrzewski	N/A	N/A
9662108	12/2016	Williams	N/A	N/A
9737302	12/2016	Shelton, IV et al.	N/A	N/A
9737303	12/2016	Shelton, IV et al.	N/A	N/A
9797486	12/2016	Zergiebel et al.	N/A	N/A
2002/0025243	12/2001	Heck	N/A	N/A
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2005/0234478	12/2004	Wixey	N/A	N/A
2006/0097026	12/2005	Shelton, IV	N/A	N/A
2006/0100644	12/2005	Viola	N/A	N/A
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2006/0255090	12/2005	Milliman	227/19	A61B 17/068
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2008/0179375	12/2007	Scirica	227/176.1	A61B 17/07207
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a division of U.S. patent application Ser. No. 17/139,518, entitled “RELOAD SHAFT ASSEMBLY FOR SURGICAL STAPLER,” filed Dec. 31, 2020, currently pending, which is a division of U.S. patent application Ser. No. 15/486,227, entitled “RELOAD SHAFT ASSEMBLY FOR SURGICAL STAPLER,” filed Apr. 12, 2017, now U.S. Pat. No. 10,905,420, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/321,618, entitled “RELOAD SHAFT ASSEMBLY FOR SURGICAL STAPLER,” filed Apr. 12, 2016. The above-referenced applications are each incorporated by reference herein in their entireties.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present application relates generally to surgical occlusion instruments and, more particularly, to surgical staplers.

Description of the Related Art

(2) Surgical staplers are used to approximate or clamp tissue and to staple the clamped tissue together. As such, surgical staplers have mechanisms to ensure that tissue is properly positioned and captured and to drive staples through the tissue. As a result, this has produced, for example, multiple triggers and handles in conjunction with complex mechanisms to provide proper stapling of the clamped tissue. With these complex mechanisms, surgical staplers can have increased manufacturing burdens, as well as potential sources for device failure and confusion for the user. Thus, reliable stapling of clamped tissue without complex mechanisms is desired.

SUMMARY OF THE INVENTION

(3) In certain embodiments, a surgical stapler is provided herein. The surgical stapler comprises an elongate shaft, a jaw assembly, and a handle assembly. The elongate shaft has a proximal end and a distal end. The elongate shaft defines a longitudinal axis between the proximal end and the distal end. The jaw assembly is positioned at the distal end of the elongate shaft. The jaw assembly comprises a first jaw, a second jaw, and a plurality of staples. The jaw assembly is selectively positionable in one of a closed configuration, an open configuration, and a firing configuration. The handle assembly is positioned at the proximal end of the elongate shaft.

(4) In certain embodiments, the elongate shaft comprises a jaw assembly at the distal end thereof coupled at an articulation joint. The articulation joint can allow articulation of the jaw assembly about an articulation range. Translation of an articulation member that extends through the elongate shaft articulates the jaw assembly. The elongate shaft further comprises a drive member extending

through the elongate shaft. The drive member has a flexible segment extending through the articulation joint. A firing member is coupled to the distal end of the drive member.

(5) In certain embodiments, the jaw assembly at the distal end of the elongate shaft comprises a reload support and an anvil pivotably coupled to the reload support. A firing member having an I-beam configuration is positioned in the jaw assembly. The jaw assembly can further comprise a lockout mechanism to prevent the firing member from being advanced unless an unfired reload is positioned in the jaw assembly.

(6) In various embodiments, a shaft coupler can be positioned at the proximal end of the shaft. The shaft coupler can be configured to engage a coupler on a handle assembly in a bayonet connection. The bayonet connection simultaneously couples an articulation member, a drive member, and the elongate shaft. The coupler can further comprise a shaft identification mechanism. The coupler can further comprise a lock-in mechanism to retain the shaft assembly in connection with the handle assembly.

(7) In various embodiments, a reload assembly for a surgical stapling system is provided. The reload assembly comprises an elongate shaft, a jaw assembly, a firing member, an actuation beam, and a reload lockout mechanism. The elongate shaft has a proximal end and a distal end. The elongate shaft defines a longitudinal axis extending between the proximal end and the distal end. The jaw assembly is positioned at the distal end of the elongate shaft. The jaw assembly comprises a first jaw, and a second jaw. The first jaw comprises a reload support configured to receive a staple reload. The second jaw is pivotably coupled to the first jaw. The second jaw comprises an anvil surface. The firing member is longitudinally slidable within the jaw assembly. The actuation beam is longitudinally slidable within the elongate shaft. The actuation beam has a proximal end and a distal end. The distal end of the actuation beam is coupled to the firing member. The reload lockout mechanism comprises a lockout lever pivotally coupled to the reload support and pivotable between a locked position preventing distal movement of the actuation beam relative to the elongate shaft and an unlocked position allowing distal movement of the actuation beam relative to the elongate shaft.

(8) In various embodiments, a reload assembly for a surgical stapling system is provided. The reload assembly comprises an elongate shaft, a jaw assembly, an actuation beam, and a shaft coupler. The elongate shaft has a proximal end and a distal end and defines a longitudinal axis extending between the proximal end and the distal end. The jaw assembly is positioned at the distal end of the elongate shaft. The jaw assembly comprises a first jaw, and a second jaw. The first jaw comprises a reload support configured to receive a staple reload. The second jaw is pivotably coupled to the first jaw. The second jaw comprises an anvil surface. The actuation beam is longitudinally slidable within the elongate shaft. The actuation beam has a proximal end and a distal end. The distal end of the actuation beam is coupled to the jaw assembly. The shaft coupler is positioned at the proximal end of the elongate shaft. The shaft coupler comprises a locking member positioned therein. The locking member is radially outwardly advanceable by distal actuation of the proximal end of the actuation beam.

(9) In various embodiments, a reload assembly for a surgical stapling system is provided. The reload assembly comprises an elongate shaft, a jaw assembly, an actuation beam, and a shaft coupler. The elongate shaft has a proximal end and a distal end and defines a longitudinal axis extending between the proximal end and the distal end. The jaw assembly is positioned at the distal end of the elongate shaft. The jaw assembly comprises a first jaw and a second jaw. The first jaw comprises a reload support configured to receive a staple reload. The second jaw is pivotably coupled to the first jaw. The second jaw comprises an anvil surface. The actuation beam is longitudinally slidable within the elongate shaft. The actuation beam has a proximal end and a distal end. The distal end of the actuation beam is coupled to the jaw assembly. The shaft coupler is positioned at the proximal end of the elongate shaft. The shaft coupler is configured to removably couple to a handle assembly. The shaft coupler comprises a lockout mechanism positioned therein.

The lockout mechanism comprises a locking ring and a lockout member. The locking ring is rotatable about the longitudinal axis. The lockout member is radially outwardly advanceable by rotation of the locking ring.

(10) In various embodiments, a reload assembly for a surgical stapling system is provided. The reload assembly comprises an elongate shaft, a jaw assembly, an actuation beam, an articulation link, a support link, and an articulation latching mechanism. The elongate shaft has a proximal end and a distal end and defines a longitudinal axis extending between the proximal end and the distal end. The jaw assembly is articulably coupled to the elongate shaft at the distal end of the elongate shaft. The jaw assembly comprises a first jaw and a second jaw. The first jaw comprises a reload support configured to receive a staple reload. The second jaw is pivotably coupled to the first jaw. The second jaw comprises an anvil surface. The actuation beam is longitudinally slidable within the elongate shaft to actuate the jaw assembly. The actuation beam has a proximal end and a distal end. The articulation link is longitudinally slidable within the elongate shaft to articulate the jaw assembly relative to the elongate shaft. The articulation link has a proximal end positioned adjacent the proximal end of the elongate shaft and a distal end pivotably coupled to the jaw assembly. The support link is longitudinally slidable within the elongate shaft. The support link has a proximal end extending longitudinally to a distal end pivotably coupled to the jaw assembly. The articulation latching mechanism is positioned within the elongate shaft between the proximal end and the distal end. The articulation latching mechanism has an unlatched configuration in which the articulation link and the support link are slidable within the elongate shaft and a latched configuration wherein the articulation latching mechanism engages the articulation link and the support link to prevent longitudinal sliding of the articulation link and the support link.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an embodiment of surgical stapling system with the jaws in an open configuration;
- (2) FIG. 2a is a perspective view of several embodiments of shaft assembly for the surgical stapling system of FIG. 1;
- (3) FIG. 2b is a perspective view of several embodiments of jaw assembly for the surgical stapling system of FIG. 1;
- (4) FIG. 3 is a perspective view of a jaw assembly at the distal end of the shaft assembly for the surgical stapling system of FIG. 1;
- (5) FIG. 4 is a perspective view of the jaw assembly of FIG. 3 with a staple reload;
- (6) FIG. 5 is a perspective view of the jaw assembly of FIG. 3 with a staple reload inserted;
- (7) FIG. 6A is a top view of an anvil for the jaw assembly of FIG. 3;
- (8) FIG. 6B is a top view of an anvil plate for the jaw assembly of FIG. 3;
- (9) FIG. 7 is an exploded perspective view of the anvil of the jaw assembly of FIG. 3;
- (10) FIG. 8 is a perspective view of the top jaw of the jaw assembly of FIG. 3 in an initial state and a formed state;
- (11) FIG. 9 is a perspective view of the anvil surface of the jaw assembly of FIG. 3;
- (12) FIG. 10 schematic diagram of staple recesses in the anvil surface of FIG. 9;
- (13) FIG. 11A is a perspective view of the anvil of the jaw assembly of FIG. 3;
- (14) FIG. 11B is a top view of the anvil of the jaw assembly of FIG. 3;
- (15) FIG. 12 is a perspective view of the reload support of the jaw assembly of FIG. 3 with a reload partially inserted;
- (16) FIG. 13 is a perspective view of the reload support of the jaw assembly of FIG. 3 with a reload inserted;

(17) FIG. 14 is a side view of a closure beam of the jaw assembly of FIG. 3;

(18) FIG. 15 is a partial cut-away front view of the closure beam of FIG. 14 with a flange thereof positioned in a channel in the anvil of the jaw assembly of FIG. 3;

(19) FIG. 16 is an exploded perspective view of a reload for use in the staple system of FIG. 1;

(20) FIG. 17 is an upper perspective view of the reload for use in the staple system of FIG. 1;

(21) FIG. 18 is a lower perspective view of the reload for use in the staple system of FIG. 1;

(22) FIG. 19 is an exploded lower perspective view of the reload for use in the staple system of FIG. 1;

(23) FIG. 20 is a perspective view of the reload for use in the staple system of FIG. 1;

(24) FIG. 21 is a top detail view of the reload for use in the staple system of FIG. 1;

(25) FIG. 22 is a perspective view of a staple pusher for the reload of FIG. 16;

(26) FIG. 23 is a perspective view of a staple pusher of the reload of FIG. 16;

(27) FIG. 24 is a perspective view of the reload for use in the staple system of FIG. 1;

(28) FIG. 25 is a partial cut-away view of the jaw assembly of FIG. 3 in a closed configuration;

(29) FIG. 26 is a perspective view of the reload of FIG. 16;

(30) FIG. 27 is a perspective view of the jaw assembly of FIG. 3 in a closed configuration with a reload inserted;

(31) FIG. 28 is a perspective view of the jaw assembly of FIG. 3 with a reload positioned for insertion;

(32) FIG. 29 is a perspective view of the jaw assembly of FIG. 3 with a reload inserted;

(33) FIG. 30 is a top view of the reload of FIG. 16;

(34) FIG. 31 is a side view of the jaw assembly of FIG. 3 with a reload inserted;

(35) FIG. 32A is a perspective view of a reload lockout mechanism of the shaft assembly;

(36) FIG. 32B is a side view of the reload lockout mechanism of the shaft assembly;

(37) FIG. 33 is a side view of the reload lockout mechanism of the shaft assembly in a locked configuration;

(38) FIG. 34 is a side view of the reload lockout mechanism of the shaft assembly in an unlocked configuration;

(39) FIG. 35 is a perspective view of the distal end of the elongate shaft at an articulation joint connection with the jaw assembly of FIG. 3;

(40) FIG. 36 is a partial cut-away perspective view of one embodiment of articulation joint at the distal end of the elongate shaft;

(41) FIG. 37 is a partial cut-away perspective view of one embodiment of articulation joint at the distal end of the elongate shaft;

(42) FIG. 38A is a partial cut-away top view of the articulation joint of FIG. 36 in an articulated position;

(43) FIG. 38B is a partial cut-away top view of the articulation joint of FIG. 36 in another articulated position;

(44) FIG. 39 is a partial cut-away perspective view of another embodiment of articulation joint at the distal end of the elongate shaft;

(45) FIG. 40 is a partial cut-away perspective view of the embodiment of articulation joint of FIG. 39 at the distal end of the elongate shaft;

(46) FIG. 41 is a partial cut-away top view of the articulation joint of FIG. 39 in an articulated position;

(47) FIG. 42 is a partial cut-away top view of the articulation joint of FIG. 39 in another articulated position;

(48) FIG. 43 is a partial cut-away top view of the articulation joint of FIG. 39 in a latched position;

(49) FIG. 44 is a side view of the proximal end of the shaft assembly positioned adjacent a handle assembly for the stapler system of FIG. 1;

(50) FIGS. 45A-45D are perspective views of a coupling of the proximal end of the shaft assembly

to the handle assembly in a stapler system of FIG. 1;

(51) FIG. 46 is an exploded perspective view of the proximal end of the shaft assembly of the stapler system of FIG. 1;

(52) FIG. 47 is a cut-away side view of the proximal end of the shaft assembly positioned adjacent a handle assembly for the stapler system of FIG. 1;

(53) FIGS. 48A-48B are perspective views of a coupling of the proximal end of the shaft assembly to the handle assembly in a stapler system of FIG. 1;

(54) FIGS. 49A-49B are perspective partial cut-away views of a coupling of the proximal end of the shaft assembly to the handle assembly in a stapler system of FIG. 1;

(55) FIG. 50 is a perspective partial cut-away view of the proximal end of the shaft assembly in a stapler system of FIG. 1;

(56) FIG. 51 is a perspective partial cut-away view of the proximal end of the shaft assembly in a stapler system of FIG. 1; and

(57) FIG. 52 is an exploded perspective view of the proximal end of the shaft assembly in a stapler system of FIG. 1.

DETAILED DESCRIPTION OF THE INVENTION

(58) With reference to FIG. 1, an embodiment of surgical stapling system is illustrated. The illustrated embodiment of surgical stapler 10 comprises an elongate shaft 20, a jaw assembly 30, and a handle assembly 40. FIG. 1 illustrates the surgical stapler 10 with the jaw assembly 30 in an open configuration. A staple reload 50 can be positioned in the jaw assembly. While the illustrated surgical stapling system is illustrated with a powered handle, it is contemplated that the elongate shaft 20 and jaw assembly 30 can be interchangeably used in a stapling system including a mechanical stapler handle. For example, it is contemplated that the various embodiments of elongate shaft assembly 20 and jaw assembly 20 described herein can be used interchangeably with either the powered handle assemblies described in U.S. patent application Ser. No. 15/486,008, entitled "SURGICAL STAPLER HAVING A POWERED HANDLE," filed Apr. 12, 2017, currently pending, and the mechanical manually actuated handle assemblies described in U.S. patent application Ser. No. 15/485,620, entitled "SURGICAL STAPLER HAVING ARTICULATION MECHANISM," filed Apr. 12, 2017, currently pending. These applications are incorporated by reference herein in their entireties.

(59) With continued reference to FIG. 1, the illustrated embodiment of surgical stapler 10 can be sized and configured for use in laparoscopic surgical procedures. For example, the elongate shaft 20 and jaw assembly 30 can be sized and configured to be introduced into a surgical field through an access port or trocar cannula. In some embodiments, the elongate shaft 20 and jaw assembly 30 can be sized and configured to be inserted through a trocar cannula having a relatively small working channel diameter, such as, for example, less than 8 mm. In other embodiments, elongate shaft 20 and jaw assembly 30 can be sized and configured to be inserted through a trocar cannula having a larger working channel diameter, such as, for example, 10 mm, 11 mm, 12 mm, or 15 mm. In other embodiments, it is contemplated that certain aspects of the surgical staplers described herein can be incorporated into a surgical stapling device for use in open surgical procedures.

(60) With continued reference to FIG. 1, as illustrated, the elongate shaft 20 comprises a generally tubular member. The elongate shaft 20 extends from a proximal end to a distal end. The elongate shaft 20 defines a central longitudinal axis, L, of the surgical stapler 10 extending between the proximal end and the distal end.

(61) With reference to FIG. 2a, it is contemplated that the stapling system can include an elongate shaft having a desired length. While the features of the jaw assembly and handle coupling described herein can be substantially similar for each of these shaft assemblies, the shaft bodies can be scalable. For example, a stapling system can include a relatively short elongate shaft 20', a mid-length elongate shaft 20, or a relatively long elongate shaft 20". Each of these shaft lengths can have particular applicability for a subset of patients or procedures. For example, the short elongate

shaft **20'** can be useful in pediatric procedures, and the long elongate shaft **20"** can be useful in bariatric procedures.

(62) With reference to FIG. **2b**, it is contemplated that the stapling system can include a jaw assembly having a desired length. While the features of the jaw assembly and articulation joint described herein can be substantially similar for each of these shaft assemblies, the jaw assemblies bodies can be scalable. For example, a stapling system can include a relatively short jaw assembly **30'**, a mid-length jaw assembly **30**, or a relatively long jaw assembly **30"**. Each of these jaw assemblies can have particular applicability for a subset of patients or procedures. In certain embodiments, it is contemplated that the jaw assembly have a length of approximately 45 mm. In other embodiments, it is contemplated that the jaw assembly have a length of approximately 60 mm.

(63) With continued reference to FIG. **1**, in the illustrated embodiment, the jaw assembly **30** is coupled to the elongate shaft **20** at the distal end **24** of the elongate shaft **20**. The jaw assembly **30** comprises a first jaw **32** and a second jaw **34** pivotally coupled to the first jaw **32**. In the illustrated embodiment, the jaw assembly **30** is articulable with respect to the elongate shaft **20**.

(64) With continued reference to FIG. **1**, in the illustrated embodiment, the jaw assembly **30** can be actuated from an open configuration (FIG. **1**) to a closed configuration to a stapling configuration by an actuation member or beam that is longitudinally slidable within the elongate shaft. In an initial position, the beam can be positioned at the distal end of the elongate shaft **20**. With the beam in the initial position, the second jaw **34** is pivoted away from the first jaw **32** such that the jaw assembly **30** is in the open configuration. The actuation beam engages the second jaw **34** upon translation of the actuation member or beam distally along the longitudinal axis **L**. Translation of the actuation beam distally from the initial position a first distance can actuate the jaw assembly from the open configuration to the closed configuration. With the jaw assembly **30** in the closed configuration, the actuation beam can be returned proximally the first distance to return the jaw assembly **30** to the open configuration. A distal end of the actuation beam can advance a staple slider configured to deploy staples from the first jaw **32** such that further translation of the actuation beam distally past the first distance deploys the plurality of staples from the reload positioned in the first jaw **32**.

(65) With continued reference to FIG. **1**, in the illustrated embodiment, the handle assembly is coupled to the elongate shaft **20** at the proximal end of the elongate shaft **20**. As illustrated, the handle assembly **40** has a pistol grip configuration with a housing defining a stationary handle **42** and a movable handle **44** or trigger pivotally coupled to the stationary handle **42**. It is contemplated that in other embodiments, surgical stapler devices including aspects described herein can have handle assemblies with other configuration such as, for example, scissors-grip configurations, or in-line configurations. As further described in greater detail below, the handle assembly **40** houses an actuation mechanism configured to selectively advance an actuation shaft responsive to movement of the movable handle **44**.

(66) With reference to FIG. **3**, an embodiment of jaw assembly at the distal end of the shaft assembly **20** is illustrated. In the illustrated embodiment, the jaw assembly comprises a reload support **210** articulably coupled to the distal end of the shaft assembly **20** at an articulation joint **230**. An anvil **220** is pivotally coupled to the reload support **210** and defines a top jaw of the jaw assembly **30**. A firing member **240** can slide within the jaw assembly to initially close the anvil **220** relative to the reload support **210**, then fire staples from a reload. In some embodiments, the firing member **240** has an I-beam configuration with a vertical beam **242** spanning between two horizontally-protruding flanges **244**, **246**. Advantageously, with an I-beam configuration, one horizontal flange **244** can engage a channel in the anvil **220** and the other flange **246** can engage a channel in the reload or reload support to close the jaw assembly then maintain a desired closed spacing of the jaw assembly when the firing member is advanced distally. In some embodiments, the firing member **240** can comprise a cutting blade **248** formed on or mounted to the vertical beam

in an I-beam configuration. This cutting blade can separate tissue as staples are fired to form staple lines on both sides of the separated tissue.

(67) With reference to FIGS. **4** and **5**, the reload support **210** can be sized to receive and retain a disposable reload **50**. The reload **50** can be lowered and moved proximally into the reload support **210** until mating features on the reload engage corresponding features on the reload support **210**.

(68) With reference to FIGS. **6A**, **6B**, **7**, and **8**, various aspects of the anvil **220** of the jaw assembly **30** are illustrated. In certain embodiments, the anvil **220** comprises an anvil plate **222** coupled to a top surface **224**. The anvil plate can comprise a longitudinal channel **225** formed therein in which a horizontal flange of the firing member rides and a longitudinal slot **227** formed through the longitudinal channel **225** in which the vertical beam of the firing member rides. The top surface **224** can be formed of a sheet of material that is subsequently formed to overly the anvil plate. (FIG. **8** illustrates the flat sheet **224'** and shaped top surface **224**). Advantageously, the addition of the top surface **224** to the anvil plate **222** enhances the strength of the anvil of the jaw assembly.

(69) With reference to FIGS. **9** and **10**, various aspects of the anvil plate **222** of the jaw assembly **30** are illustrated. The anvil plate comprises a plurality of staple forming pockets **223** thereon. In the illustrated embodiment, the staple forming pockets **223** are positioned in two arrays of three rows with the arrays positioned on either side of the slot for the firing member. Thus, the stapler can form two sets of three linear rows of staples with the sets separated by divided tissue. In other embodiments, it is contemplated that the anvil can include staple forming pockets configured to form other numbers and configurations of staples. The staple forming pockets have a tapered configuration with a relatively large staple entry side narrowing to a relatively small staple formation side. Advantageously, this tapered configuration can guide staples to complete formation and reduce the incidence of poorly formed staples. Adjacent rows of staples can be longitudinally offset from one another such that the relatively wide entry sides of all of the rows are offset from one another to reduce the overall width of the sets of staple rows.

(70) With reference to FIGS. **11a** and **11b**, in certain embodiments of anvil **220**, the top surface **224** can be coupled to the anvil plate **222** by a welding operation along a weld line **226**. Advantageously, this closed anvil formed by the welding operation covers the channel for the firing member.

(71) With reference to FIGS. **12** and **13**, insertion of a reload **50** in the reload support **210** is illustrated. The reload support can comprise proximal jaw tabs **212** that protrude radially inwardly from side walls of the reload support **210** adjacent the proximal end thereof. The reload can comprise a relatively short, tapered proximal deck **510** sized to be positioned under and retained by the proximal jaw tabs. Moreover, the reload **50** can include retention tabs **512** protruding laterally outwardly adjacent a distal end thereof. The reload support **210** can comprise a corresponding pair of retention recesses **214** sized and configured to receive the retention tabs when the reload is positioned in the reload support.

(72) With reference to FIGS. **14** and **15**, an embodiment of firing member **240** having an I-beam configuration is illustrated. In the illustrated embodiment, The firing member comprises a vertical beam **242** having a cutting blade formed therein at a leading edge. The cutting blade comprises a curved cutting blade **248**. A trailing edge of the firing member **240** comprises a drive member interface **245** such as a cutout or protrusion to allow the firing member to be securely coupled with the drive member extending through the elongate shaft. The trailing edge of the firing member **240** can further comprise a lockout interface **247**, such as a proximally extending 'tail' that can position a reload lockout in an unlocked configuration when the firing member is in a proximal position. The firing member further comprises an upper horizontal flange **244** configured to ride in the channel **225** of the anvil and a lower horizontal flange **246** configured to engage the reload or reload support. As illustrated in FIG. **15**, although the firing member has a general I-beam configuration, in some embodiments the horizontal flanges are curved or tapered such to conform with a shape of the channel **225** in the anvil. In some embodiments, the firing member **240** can

further be configured to reduce friction during a firing sequence such as by surface finishing operations, addition of a film lubricant, or deposition of a low-friction surface on the firing member, channel, or both.

(73) With reference to FIGS. **16-19**, an embodiment of reload **50** for use in the stapling system is illustrated. The reload **50** comprises a plurality of staples **520** positioned in a corresponding plurality of staple pockets **532** formed in a cartridge **530**. The staple pockets **532** are arranged in two sets of three rows each with each set separated by a slot formed through the cartridge **530**. The staples **520** rest in a plurality of staple pushers **540** underlying the staple pockets **532**. A slider **550** having a ramp **552** corresponding to each row of staple pusher **540** and a lockout tail **554** is positioned at the proximal end of the reload. The slider **550** is longitudinally slidable within the reload responsive to movement of the firing member. A jacket **560** underlies the cartridge and maintains the staples and staple pushers in the staple pockets. The jacket can have protruding hooks **562** to engage the cartridge.

(74) With reference to FIG. **20**, in some embodiments, the reload **50** can include a shipping cover **570** covering an upper surface of the cartridge. Advantageously, the shipping cover **570** can prevent one or more of the staples from becoming dislodged from or misaligned within the staple pockets before the reload is used. The shipping cover **570** is removed before the reload **50** is positioned in the reload support.

(75) With reference to FIGS. **21-23**, in some embodiments, the reload **50** can include certain staple alignment and retention features. For example the staple pockets **532** formed in the cartridge **530** can include staple guides **534** at ends thereof to receive legs of the staples **520** positioned therein. The staple pushers **540** can additionally include nubs **542** sized and configured to ride in the staple guides **534**. As illustrated, in certain embodiments, the staple pushers **540** can be formed in groups of three such that one staple pusher **540** can push a single staple in each of three adjacent rows of staples. Moreover, an upper surface **544** of each of the staple pushers **540** can include a staple saddle configuration to relatively securely receive a staple. Secure positioning of the staples **520** in the staple pushers **540** and engagement of the staple legs and nubs **542** of the staple pushers with the staple guides can advantageously reduce the incidence of misaligned or malformed staples.

(76) With reference to FIGS. **24-31**, in various embodiments, the reload **50** and jaw assembly can be configured to be securely coupled to one another to align the staple pockets on the reload **50** with the staple forming pockets on the anvil and maintain the position of the reload **50** in the jaw assembly during staple firing. The reload **50** can include upwardly protruding bosses **538** at a proximal end thereof (FIG. **24**) that define a tissue gap between the anvil **220** and an upper surface of the cartridge **530** of the reload **50** with the jaw assembly in a closed configuration (FIG. **25**). Moreover, the retention tabs **512** formed adjacent the distal end of the reload (FIG. **26**) are positioned within recesses **214** of the reload support **210** and prevent the reload from shifting distally during a firing operation. Thus, the reload **50** can be rapidly and securely coupled to the reload support **210** (FIGS. **28-29**). Additionally, a proximal end of the cartridge **530** can taper to a reduced height to further facilitate placement on the reload support (FIG. **30**). Furthermore, the cartridge can be configured with a lowered distal end **514** having a profile protruding below the reload support (FIG. **31**). This lowered profile ensures secure engagement of the reload with the reload support.

(77) With reference to FIGS. **32A**, **32B**, and **33-34**, in certain embodiments, the jaw assembly can comprise a reload lockout mechanism **580**. The reload lockout mechanism **580** can prevent advancement of the firing member if no reload is positioned within the jaw assembly or if an empty reload is positioned within the jaw assembly. The reload lockout mechanism **580** includes a lockout lever **582** pivotally coupled to the reload support. An axis defined by the pivot extends generally transverse to the longitudinal axis of the elongate shaft. With the firing member **240** fully retracted such that the jaw assembly is in an open configuration, a tail **247** extending proximally from the firing member **240** maintains the lockout lever **582** pivoted to the unlocked position. In the

illustrated embodiment, a proximal portion of the lockout lever **582** proximal the pivot is forked or bifurcated to receive the firing member **240** therein such that the tail **247** can act on a surface of the lockout lever **582** distal the pivot. If no reload is inserted, an attempt to advance the firing member **240** will allow the lockout lever to pivot about a pivot point **584** from the unlocked position to the locked position as the tail **247** of the firing member is advanced distally along the lockout lever. (FIG. **33**). With the lockout lever **582** in the locked position, a proximal, locking end **586** of the lockout lever interferes with a lock recess on the drive member **26**, preventing further distal movement of the drive member.

(78) With continued reference to FIGS. **32A**, **32B**, and **33-34**, if an unfired reload is inserted into the reload support (FIG. **34**), a tail **554** extending proximally from the slider **550** engages a distal end of the lockout lever **582**. As illustrated, the tail **554** acts on a lower surface of a distal portion of the lockout lever **582** distal the pivot point. This engagement of the slider tail **554** with the distal end of the lockout lever **582** pivots the proximal end of the lookout lever **582** away from the drive member **26** even once the tail **247** of the firing member **240** is no longer acting on the proximal portion of the lockout lever. Accordingly, the drive member **26** and firing member **240** can be distally advanced to fire the staples from the reload. Upon completion of a firing stroke, the slider **550** remains at a distal end of the reload. Thus if the jaw assembly is returned to the open configuration, withdrawing the firing member, the fired reload should be removed and a new unfired reload should be inserted to unlock the reload lockout.

(79) With reference to FIGS. **35-37**, an embodiment of articulation joint **300** to couple the jaw assembly **30** to the distal end of the elongate shaft **20** is illustrated. In the illustrated embodiment, the articulation joint **300** comprises an articulation rod **310** pivotably coupled to the jaw assembly laterally offset from a central longitudinal axis of the shaft assembly. A pivot joint is positioned along the central longitudinal axis. The articulation joint **300** further comprises a support link **320** pivotably coupled to the jaw assembly laterally offset from the central longitudinal axis of the shaft and opposite the articulation rod. The drive beam **26** extends longitudinally along the central longitudinal axis between the articulation rod **310** and the support link **320**. At least a segment of the drive beam **26** extending through the articulation joint **300** is flexible. In some embodiments, the drive beam **26** can be coupled to a flexible segment comprising a stack of shim material, which is flexible while maintaining desired force transmission capabilities for a staple firing operation. The articulation joint can further comprise one or more drive member bearings **330** positioned laterally outwardly of the drive beam **26**. In some embodiments, the drive bearings **330** can comprise a flexible plastic material (FIG. **36**). In other embodiments, the drive bearings **330'** can be comprised of a metal shim material (FIG. **37**). Advantageously, the metal shim drive bearing **330'** can be keyed into the shaft to provide support to the flexible segment of the drive member. Moreover, the metal shim bearings can have a relatively low profile configuration. The metal shim bearings can include a low friction coating such as a TEFLON coating to reduce friction during a firing.

(80) With reference to FIGS. **38A-38B**, articulation of the articulation joint to position the jaw assembly in a first articulation position and a second articulation position are illustrated. The articulation rod **310** can be translated proximally (FIG. **38A**) or distally (FIG. **38B**) relative to the shaft. The lateral offset positioning of the articulation rod **310** articulates the jaw assembly relative to the shaft responsive to translation of the articulation rod. The support link **320** opposite the articulation rod **310** is passive, but can guide articulation motion of the jaw assembly and can advantageously assist in maintaining the flexible portion of the drive beam **26** towards the center of the shaft at the articulation joint, preventing the flexible portion of the drive beam **26** from buckling at the articulated bend at the articulation joint. In other embodiments, the articulation joint can include two articulation rods instead of an articulation rod and support link. In embodiments with two articulation rods, an articulation latch mechanism can be positioned in the shaft to prevent undesired articulation once a staple firing operation has commenced. For example, a latch or brake

mechanism can retain the articulation rods from further movement once the drive beam **26** is translated distally.

(81) With reference to FIGS. **39-40**, another embodiment of articulation joint **300'** to couple the jaw assembly **30** to the distal end of the elongate shaft **20** is illustrated. The articulation joint **300'** comprises an articulation latch mechanism **340** positioned in the elongate shaft. In the illustrated embodiment, the articulation joint **300'** comprises an articulation rod **310'** pivotably coupled to the jaw assembly laterally offset from a central longitudinal axis of the shaft assembly. A pivot joint is positioned along the central longitudinal axis. The articulation joint **300'** further comprises a support link **320'** pivotably coupled to the jaw assembly laterally offset from the central longitudinal axis of the shaft and opposite the articulation rod. The drive beam **26'** extends longitudinally along the central longitudinal axis between the articulation rod **310'** and the support link **320'**. At least a segment of the drive beam **26'** extending through the articulation joint **300'** is flexible. In some embodiments, the drive beam **26'** can be coupled to a flexible segment comprising a stack of shim material, which is flexible while maintaining desired force transmission capabilities for a staple firing operation. The articulation joint can further comprise one or more drive member bearings **330** positioned laterally outwardly of the drive beam **26'**. In some embodiments, the drive bearings **330** can comprise a flexible plastic material (FIG. **36**). In other embodiments, the drive bearings **330'** can be comprised of a metal shim material (FIG. **37**). Advantageously, the metal shim drive bearing **330'** can be keyed into the shaft to provide support to the flexible segment of the drive member. Moreover, the metal shim bearings can have a relatively low profile configuration. The metal shim bearings can include a low friction coating such as a TEFLON coating to reduce friction during a firing.

(82) With reference to FIGS. **41-42**, articulation of the articulation joint to position the jaw assembly in a first articulation position and a second articulation position are illustrated. The articulation rod **310'** can be translated proximally (FIG. **41**) or distally (FIG. **42**) relative to the shaft. The lateral offset positioning of the articulation rod **310'** articulates the jaw assembly relative to the shaft responsive to translation of the articulation rod. The support link **320'** opposite the articulation rod **310'** is passive, but can guide articulation motion of the jaw assembly and can advantageously assist in maintaining the flexible portion of the drive beam **26'** towards the center of the shaft at the articulation joint, preventing the flexible portion of the drive beam **26'** from buckling at the articulated bend at the articulation joint. In other embodiments, the articulation joint can include two articulation rods instead of an articulation rod and support link.

(83) With reference to FIGS. **39-43**, the articulation latch mechanism **340** or brake mechanism of the articulation joint **300'** can retain the articulation rod and support link from further movement once the drive beam **26'** is translated distally. In the illustrated embodiment, the latch mechanism **340** is positioned within the elongate shaft between the proximal end and the distal end thereof. The articulation latching mechanism **340** has an unlatched configuration in which the articulation rod and the support link are slidable within the elongate shaft. Thus, with the articulation latching mechanism in the unlatched configuration, a user can articulate the jaw assembly relative to the elongate shaft by operation of an articulation control on the handle assembly. The articulation latching mechanism **340** further comprises a latched configuration (FIG. **43**), wherein the articulation latching mechanism engages the articulation rod and the support link to prevent longitudinal sliding of the articulation link and the support link relative to the elongate shaft. Thus, in the latched configuration, the jaw assembly is retained in an articulated position and the user is prevented from articulating the jaw assembly relative to the elongate shaft.

(84) With continued reference to FIGS. **39-43**, in the illustrated embodiment, the articulation latching mechanism **340** comprises a first latch surface, such as a first plurality of teeth **342** formed on the articulation rod **310'**. As illustrated, the first plurality of teeth **342** is positioned within the elongate shaft between the proximal end and the distal end of the articulation rod **310'**. The articulation latching mechanism **340** can further comprise a second latch surface, such as a second

plurality of teeth **344** formed on the support link **320'**. As illustrated, in the embodiment of elongate shaft assembly having a latching articulation mechanism, the support link **320'** can extend proximally within the shaft through the articulation latching mechanism **340**. In the illustrated embodiment, the second plurality of teeth **344** is positioned between the proximal end of the support link and the distal end of the support link adjacent the proximal end of the support link **320'**.

(85) In the illustrated embodiment, the articulation latching mechanism **340** further comprises a first shoe **346** having a mating surface such as a first pawl surface **348** formed thereon. The first pawl surface **348** is sized and configured to be engageable with the first plurality of teeth **342**. The first shoe **346** can have a deployment surface opposite the mating surface, the deployment surface is in sliding engagement with the drive beam **26'**. The articulation mechanism **340** can further comprise a second shoe **350** having a mating surface such as a second pawl surface **352** formed thereon. The second pawl surface **352** is sized and configured to be engageable with the second plurality of teeth **344**. The second shoe **350** can have a deployment surface opposite the mating surface, the deployment surface in sliding engagement with the drive beam **26'**. The articulation latching mechanism **340** can further comprise a latching profile formed on the drive beam **26'** between the proximal end and the distal end thereof and positioned within the elongate shaft. In the illustrated embodiment, the drive beam **26'** comprises a recess segment **360** formed therein, a tapered or ramped segment **362** proximal the recess segment, and a latching segment **364** proximal the ramped segment. The recess segment **360** has a first width in a direction generally perpendicular to the longitudinal axis of the elongate shaft, and the latching segment **364** has a second width greater than the first width. The articulation latching mechanism can further comprise a biasing member such as a spring clip **370** coupled to the first and second shoes and biasing the shoes **346**, **350** out of engagement with the first and second pluralities of teeth **342**, **344**. The spring clip can also maintain engagement of the deployment surfaces of the shoes **346**, **350** with the latching profile of the drive beam **26'**.

(86) With continued reference to FIGS. **39-43**, in operation, the articulation latching mechanism **340** can initially be positioned in the unlatched configuration (FIGS. **39-42**) such that the jaw assembly can be articulated to a desired orientation relative to the elongate shaft. In this initial positioning, the drive beam **26'** is in a proximal position relative to the elongate shaft, corresponding to an open or partially closed configuration of the jaw assembly. In the unlatched configuration, the first and second shoes **346**, **350** are positioned adjacent the recess segment **360** of the drive beam **26'** in a radially inward position. Once a desired articulated position of the jaw assembly has been selected, a user can proceed to close and fire the jaw assembly, resulting in distal actuation of the drive beam **26'** relative to the elongate shaft. This distal movement of the drive beam **26'** advances the ramped and latching segments **362**, **364** over the deployment surfaces of the first and second shoes **346**, **350**, advancing the shoes radially outwardly. (FIG. **43**). With the first and second shoes **346**, **350** in the radially outward configuration, the first pawl surface **348** engages the first plurality of teeth **342**, and the second pawl surface **348** engages the second plurality of teeth **342** to configure the articulation latch mechanism in the latched configuration. Opening the jaw assembly after a firing sequence will reverse the sequence and return the articulation latch to the unlatched configuration. Thus, desirably, actuation of the drive member **26'** to close and fire the jaw assembly automatically latches an articulated position of the jaw assembly. Advantageously, this latching can reduce or prevent any tendency of the jaw to 'wag' relative to the elongate shaft as the drive beam is advanced around and retracted through the articulation bend. While the illustrated embodiment of actuation latching mechanism includes meshing arrays of teeth on the shoes and actuation rod and support link that define a plurality of discreet latched positions, it is contemplated that in other embodiments, the shoes, actuation rod, and support link can be configured to frictionally engage to define a continuous array of latched articulation positions. Moreover, while the illustrated embodiment includes two shoes each engageable with a

corresponding plurality of teeth, in other embodiments, a single shoe can be advanceable to engage a single plurality of teeth on the articulation rod or support link.

(87) With reference to FIGS. **44** and **45A-45D**, a coupler **46** at the distal end of the handle assembly **40** can be coupled to the proximal end of the shaft assembly **20**. The coupler **46** can include a bayonet connection with a lock-in. In the illustrated embodiment, the reload shaft **20** to handle **40** connection comprises a bayonet style connection, in which a user axially aligns and inserts the reload shaft **20** into the handle **40** and rotates the reload shaft **20** approximately 90 degrees to connect. This bayonet connection operatively couples two mechanical functions of the reload shaft **20** to corresponding actuators of the handle **40**. When the bayonet connection is fully coupled, an articulation member within the shaft **20** is coupled to an articulation adapter of the handle and a drive member within the shaft **20** is coupled to the actuation adapter. Furthermore, the handle **40** and shaft **20** can be configured with a latch mechanism at the coupler **46** to prevent a user from removing the shaft **20** once the actuation adapter and drive member has been activated. Moreover, the connection at the coupler **46** can include a reload identifying mechanism such that the control system of the handle can detect if a reload shaft is connected, and if so what the attached jaw length of the reload is. It is contemplated that the handle can be used with reload shafts **20** including different length jaw assemblies. In some embodiments the same handle **40** can be used with either 45 mm or 60 mm length jaw assemblies.

(88) In FIG. **45A**, the shaft **20** is positioned in alignment with the coupler **46** on the handle, and a release knob of the coupler **46** is withdrawn to expose a bayonet channel **152** of the coupler **46** on a rotation insert of the coupler **46**. The shaft **20** can include a retention post **22** or boss positionable within the bayonet channel **152**. In the illustrated embodiment, the shaft includes two bosses positioned 180 degrees apart on the outer surface thereof and the coupler **46** includes a corresponding two bayonet channels **152**. It is contemplated that in other embodiments, other numbers and configurations of bosses and bayonet channels can be used to provide a desired connection strength and ease of alignment.

(89) With reference to FIG. **45B**, the retention post **22** of the shaft is positioned within the bayonet channel **152**. With reference to FIG. **45C**, the reload shaft **20** has been rotated 90 degrees relative to the handle such that the retention post **22** of the shaft has reached a connected end of the bayonet channel **152**. With reference to FIG. **45D**, the release knob of the coupler is released to allow a retention recess **154** on the release knob to retain the retention post **22** of the reload shaft **20**.

(90) With reference to FIGS. **46** and **47**, the shaft assembly can include a tubular shaft with the drive member or drive beam **26** and articulation member **206** extending therethrough from the proximal end to the distal end. The drive member can extend generally centrally through the shaft assembly while the articulation member is laterally offset. The proximal end of the tubular shaft can include a coupling collar **410** for coupling to the coupler **46** at the distal end of the handle. In the illustrated embodiment, the shaft assembly can include a proximal shaft 'lock out' mechanism. The lockout mechanism comprises a locking ring positioned within a shaft coupler at the proximal end of the elongate shaft and at least one lockout member radially outwardly advanceable through the coupling collar **410**. The lockout member can be biased radially outwardly, but held in a radially inward position by the locking ring in an initial position. When the proximal end of the shaft is coupled to a handle assembly in a rotation sequence corresponding to a bayonet connection, the locking ring is engaged with a mating surface in the handle assembly and rotates relative to the elongate shaft. This rotation of the locking ring releases the lockout member. Upon removal of the shaft from the handle assembly, the lockout member radially expands. In this expanded position, the lockout member interferes with recoupling the elongate shaft to the handle assembly. Thus, this lockout mechanism can serve to limit inadvertent reuse of an elongate shaft assembly.

(91) With reference to FIGS. **48A** and **48B**, engagement of the bayonet coupling between the shaft assembly and the handle is illustrated. The coupler of the handle can comprise a rotation sleeve for coupling to the coupling collar **410** in which an actuation adapter **124**, an articulation adapter **204**,

and an identification sleeve **208** are positioned. During a bayonet coupling, the drive member of the shaft engages **26** with the actuation adapter **124**, the articulation member **206** of the shaft engages with the articulation adapter **204**, and a shaft identifier engages with the identification sleeve **208**. FIGS. **49A** and **49B** illustrate the respective engagements with the shaft in a coupled configuration.

(92) With reference to FIGS. **50** and **51**, instead of or in addition to the lockout mechanism described with reference to FIGS. **46** and **47**, certain embodiments of elongate shaft can include a lock-in or retention mechanism that operates upon initial distal advancement of the actuation adapter **124**. As illustrated, a locking member **24** is pivotably coupled to a proximal end of the shaft **20**. The locking member **24** can include a ramped or tapered lock surface at a proximal edge thereof. As illustrated in FIG. **50**, the shaft **20** is in a coupled, but unlocked configuration with respect to the coupler **46**. In the coupled, unlocked configuration, the shaft **20** can be removed from the coupler **46** through the bayonet connection by a reverse of the sequence of operations of FIGS. **45A-45D**. Once the actuation adapter **124** is advancing to operate the stapler, the actuation adapter **124** interacts with the ramped surface of the locking member **24** to advance the locking member radially outward into a locked position. In the locked position (FIG. **51**), the locking member **24** engages a locking ledge on the coupler **46** to lock in the shaft. With the shaft **20** locked in with respect to the handle **40**, the shaft **20** cannot be removed from the handle **40** until the actuation adapter **124** has been returned to a fully proximally retracted position (typically corresponding to a return to a jaws open configuration following a full closure and stapling cycle of the jaw assembly).

(93) Thus, the “lock In” feature prevents a user from removing the shaft from the handle once the drive member **26** has been driven forward. Once the locking member **24** is situated in the slot or ledge of a rotation insert of the coupler **46**, a release knob of the coupler **46** is restricted from being pulled back. This locking action on the coupler prevents the user from rotating the shaft **20** out of the bayonet connection of the coupler **46**.

(94) With reference to FIG. **52**, a proximal end of the shaft assembly comprises a shaft coupler or coupling collar **410** positioned on the proximal end of the tubular shaft. Thus, the stapling system described herein can easily be adapted for use with shaft assemblies having various diameters. In some embodiments, an inner diameter the shaft coupler can be readily resized to accommodate various tubular shafts without requiring different handle assemblies to accommodate shaft assemblies of various diameters.

(95) Although this application discloses certain preferred embodiments and examples, it will be understood by those skilled in the art that the present inventions extend beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. Further, the various features of these inventions can be used alone, or in combination with other features of these inventions other than as expressly described above. Thus, it is intended that the scope of the present inventions herein disclosed should not be limited by the particular disclosed embodiments described above, but should be determined only by a fair reading of the claims.

Claims

1. A reload assembly for a surgical stapling system, the reload assembly comprising: an elongate shaft having a proximal end and a distal end and defining a longitudinal axis extending between the proximal end and the distal end; a jaw assembly positioned at the distal end of the elongate shaft, the jaw assembly comprising: a first jaw comprising a reload support configured to receive a staple reload; and a second jaw pivotably coupled to the first jaw, the second jaw comprising an anvil surface; an actuation beam longitudinally slidable within the elongate shaft, the actuation beam having a proximal end and a distal end, the distal end of the actuation beam coupled to the jaw assembly; and a shaft coupler at the proximal end of the elongate shaft, the shaft coupler comprising a locking member positioned therein, a proximal portion of the locking member

radially outwardly advanceable relative to the longitudinal axis by distal actuation of the proximal end of the actuation beam to place the shaft coupler in a locked configuration; wherein the shaft coupler has an unlocked configuration with the locking member in a radially inward position, the shaft removably positionable in a coupler of a handle assembly in the unlocked configuration; and wherein with the shaft coupler in the locked configuration and the shaft positioned in the coupler of the handle assembly, the shaft is locked in with respect to the handle assembly.

2. The reload assembly of claim 1, wherein with the shaft coupler in the locked configuration the proximal portion of the locking member is radially outwardly advanced.

3. The reload assembly of claim 1, wherein the locking member is pivotably coupled to the elongate shaft and pivotable from an unlocked position to a locked position.

4. The reload assembly of claim 1, wherein the locking member comprises a lever having a proximal end, the locking member comprising a tapered lock surface at the proximal end that is radially outwardly advanceable relative to the longitudinal axis by distal actuation of the actuation beam.

5. The reload assembly of claim 4, wherein proximal retraction of the actuation beam to a fully proximally retracted position returns the shaft coupler to the unlocked configuration.

6. The reload assembly of claim 1, wherein the shaft coupler is configured to couple to the coupler of the handle assembly in a bayonet connection and wherein with the shaft coupler in the locked configuration, a decoupling operation of the bayonet configuration is prevented.

7. The reload assembly of claim 1, wherein the locking member is outwardly advanceable upon initial distal movement of the actuation beam from a fully proximally retracted position.

8. The reload assembly of claim 7, wherein with the actuation beam in the fully proximally retracted position, the jaw assembly is in an open configuration, and wherein the actuation beam is distally longitudinally advanceable a first distance from the fully proximally retracted position to actuate the jaw assembly to a closed configuration.

9. The reload assembly of claim 8, wherein distal longitudinal advancement of the actuation beam distally past the first distance deploys staples from the staple reload received in the reload support.
